TELECOM ParisTech

SD-TSIA205

Exercise 2

Density estimation

Let X be a random variable in $[0,1]$. We assume that it admits a density denoted by f. The purpose of this problem is to estimate f given samples from X. We assume that $f \in \mathcal{C}^2([0,1])$.
Consider N independently and identically distributed samples $x_1, \dots, x_N$. We cut $[0,1]$ into m equal boxes $B_1, \dots, B_m$ of size $h := \frac{1}{m}$ where :

$$B_j = \left[\frac{j-1}{m}, \frac{j}{m}\right)$$

For each $x \in [0,1]$, we consider approximating $f(x)$ using a count of the samples in its associated box $\widehat{p}_j$ defined by :

$$\widehat{p}_j := \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}_{B_j}(x_n), \quad \widehat{f_N}(x) := \sum_{j=1}^{m} \frac{\widehat{p}_j}{h}\mathbf{1}_{B_j}(x)$$

1. Compute the bias and the variance of the estimator $\widehat{p}_j$ as a function of $p_j := \int_{B_j} f(x)dx$.
2. Let $x \in B_j$. Show that :
$$\mathbb{E}[\widehat{f_N}(x)] = \frac{p_j}{h}, \quad \mathbb{V}(\widehat{f_N}(x)) = \frac{p_j(1-p_j)}{nh^2}$$
with $\mathbb{V}(Z) = \mathbb{E}[(Z - \mathbb{E}[Z])^2]$ for any random variable Z.
3. For $x \in B_j$, show that the bias of the estimator can be written as :
$$b(x) := \mathbb{E}[\widehat{f_N}(x)] - f(x) = f'(x)\left((j - \frac{1}{2})h - x\right) + \mathcal{O}(h^2)$$
4. Let m_j be the mid-point of B_j given by $(j - \frac{1}{2})h$. Show that :
$$f'(x) = f'(m_j) + \mathcal{O}(h).$$
5. Conclude :
$$\int_0^1 b^2(x)dx = \frac{h^2}{12}\int_0^1 f'^2(x)dx + o(h^2).$$
6. Show that :
$$\int_0^1 \mathbb{V}(\widehat{f_n}(x))dx = \frac{1}{nh} + \mathcal{O}(\frac{1}{n})$$
7. What happens to the bias and to the variance as $h \to 0$?
8. Deduce that the mean integrated squared error (MISE) of the estimator verifies :
$$\mathcal{R}(\widehat{f_n}, f) := \mathbb{E}\left[\int_0^1 (\widehat{f_n}(x) - f(x))^2\right]dx = \frac{h^2}{12}\int_0^1 f'^2(x)dx + \frac{1}{nh} + o(h^2) + o\left(\frac{1}{nh}\right)$$
9. Suggest a theoretical target $h^\star$.